

Mohamed Irfan M Y

Software, IoT Application Developer & IT Specialist

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LinkedIn — Portfolio — GitHub

Career Summary

Electronics and Communication Engineering student with experience in software development, IoT systems, cybersecurity tools, web and mobile applications, and cloud integration. Skilled in Flutter, JavaScript, Python, Firebase, embedded systems, and Android security analysis.

Education

Kalasalingam University

B.Tech in Electronics and Communication Engineering

2022 – 2026

Technical Skills

- **Programming:** C, Embedded C, JavaScript, Python, Flask
- **Frontend & App Dev:** HTML, CSS, JavaScript, Flutter
- **Backend / Cloud:** Flask REST API, Firebase (Auth, Database, Storage)
- **IoT & Embedded:** ESP32, STM32, sensors, real-time systems
- **Cybersecurity:** Android static analysis, APK reverse engineering, vulnerability detection
- **Tools:** Git/GitHub, debugging, system integration
- **Networking:** Basic networking, security fundamentals

Experience

Project Intern – Microcontroller Firmware & PCB Design

C-DAC, Bangalore

- Developed embedded firmware in C for sensor data acquisition and processing.
- Implemented ADC, GPIO, UART, I2C, SPI drivers for real-time communication.
- Performed PCB bring-up, debugging and hardware validation.
- Supported hardware–software integration and testing.
- Assisted in schematic design, PCB review and fabrication workflow.

Projects

HEAL Lite – Tremor Monitoring System with AI

Dec 2025

- Built wearable-based tremor monitoring and visualization system.
- Developed Flutter app for real-time and historical analysis.
- Integrated Firebase for authentication and cloud sync.
- Designed hybrid storage with offline logging support.
- Added AI-based tremor trend analysis.

APKShield – Android Security Analysis Platform

Apr 2026

GitHub Repository

- Built web-based APK static analysis tool for vulnerability detection.
- Extracted AndroidManifest.xml and analyzed 34+ dangerous permissions.
- Created 20+ regex rules for secrets like API keys and tokens.
- Designed 4-layer binary XML decoder engine.
- Built Flask backend with single-page dashboard and threat scoring.
- Generated automated security reports in seconds.

Achievements

Runner-Up – IoTropolis (VINTRA'24)

Oct 2024

- Developed IoT gas leakage detection and alert system with cloud notification.

Certifications

- **Fortinet Certified Associate (FCA)** – Fortinet 2026
ID: 8816711595MI
Verify
- **Networking Basics** – Cisco 2026
Credential
- **One Million Prompters** – Dubai Future Foundation 2026
Certificate
- **Cybersecurity Training** – Infoziant 2026
- **Human Resources: Payroll** – LinkedIn Learning 2025
- **Wadhvani Ignite India** – WF Global 2025
Certificate